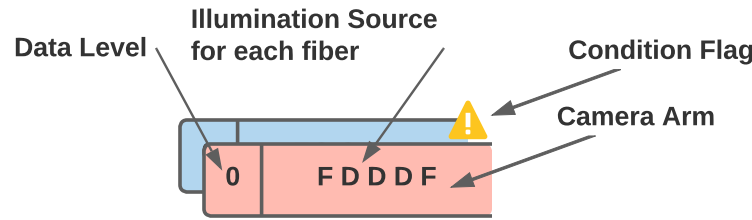
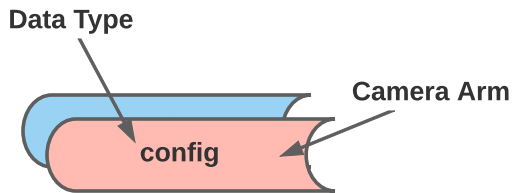


Glossary

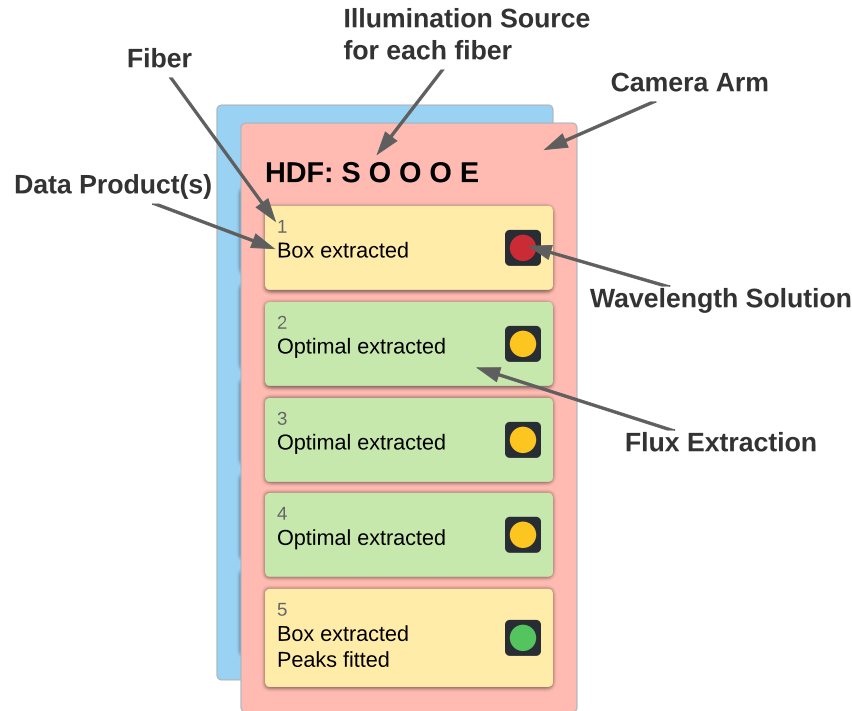
2D FITS Files:



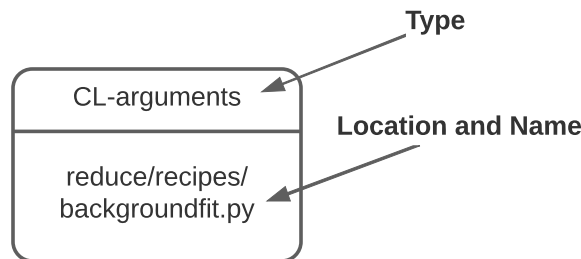
Static Input Files:



HDF Files:



Scripts/Routines:



Data Level:

- 0: Raw data from CCD
- 1: Intermediate 2D data product
- 2: Final 2D data product



Illumination Source:

- F: Flat
- D: Dark
- T: ThAr
- E: Etalon
- I: Iodine
- U: Unknown
- S: Sky
- O: Object
- X: LFC



Camera Arm:

- Blue: Blue arm (490 - 670nm)
- Red: Red arm (650 - 920 nm)



Condition Flag:

Flag indicates conditions to be met, i.e. specific exposure time, ND filter setting, etc.



Data Type:

- Config: HDF file with static instrument config information (order locations, order and fiber identification, bad pixel mask, static wavelength solution, etc.)
- Exposuremeter: Pickle file with exposuremeter data from database
- Etalon params: HDF file with etalon parameters (cavity length and chromatic terms)



Fiber:

- 1-5: Physical fiber
- 6: Virtual fiber (combines 2-4)



Flux Extraction:

- yellow: Box extraction only
- green: Box and optimal extraction



Wavelength Solution:

- red: Static - from config file
- yellow: Simultaneous - from simcal fiber and etalon calibration frame
- green: In place - from etalon source in same fiber and frame



Data Products:

- + Box extracted
- + Optimal extracted (incl. box extraction)
- + Peaks fitted (Etalon lines only)



Type:

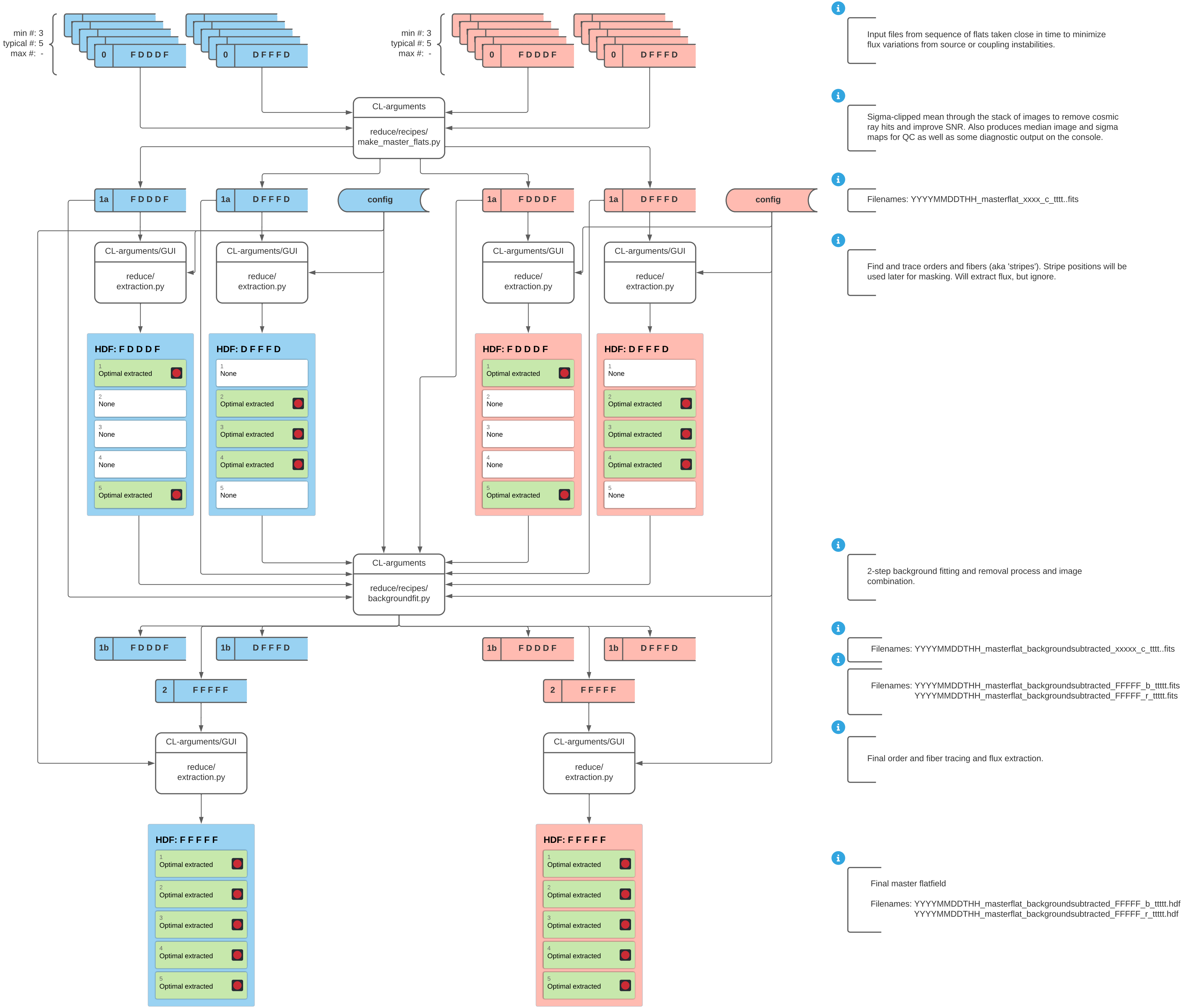
- CL-arguments: Python script accepting command-line arguments for specify I/O and options.
- GUI: Graphical User Interface
- Batch Wrapper: Python script providing a wrapper to support multi-processing batch handling while accepting CL arguments.



Location and Name:

Location within Repo and name of python file.

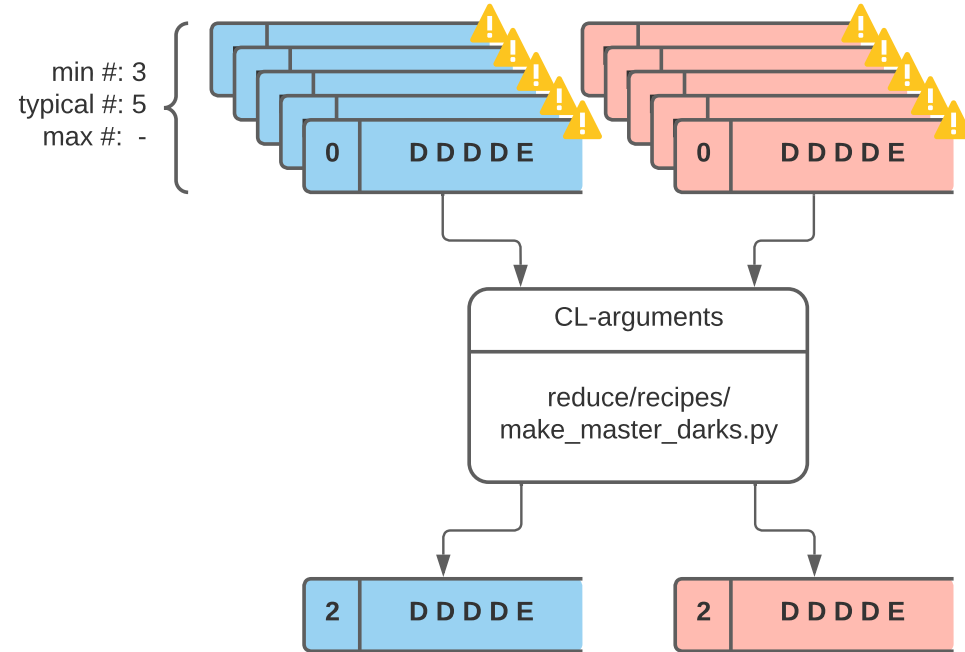
Master Flats



- Input files from sequence of flats taken close in time to minimize flux variations from source or coupling instabilities.
- Sigma-clipped mean through the stack of images to remove cosmic ray hits and improve SNR. Also produces median image and sigma maps for QC as well as some diagnostic output on the console.
- Filenames: YYYYMMDDTHH_masterflat_xxxx_c_tttt.fits
- Find and trace orders and fibers (aka 'stripes'). Stripe positions will be used later for masking. Will extract flux, but ignore.
- 2-step background fitting and removal process and image combination.
- Filenames: YYYYMMDDTHH_masterflat_backgroundsubtracted_xxxx_c_tttt.fits
- Filenames: YYYYMMDDTHH_masterflat_backgroundsubtracted_FFFFF_b_tttt.fits
YYYYMMDDTHH_masterflat_backgroundsubtracted_FFFFF_r_tttt.fits
- Final order and fiber tracing and flux extraction.
- Final master flatfield

Filenames: YYYYMMDDTHH_masterflat_backgroundsubtracted_FFFFF_b_tttt.hdf
YYYYMMDDTHH_masterflat_backgroundsubtracted_FFFFF_r_tttt.hdf

Master Darks



i

Input files from sequence of darks+etalon taken close in time to minimize flux variations from source or coupling instabilities. Files must match in exposure time and ND filter setting on a per-arm basis.

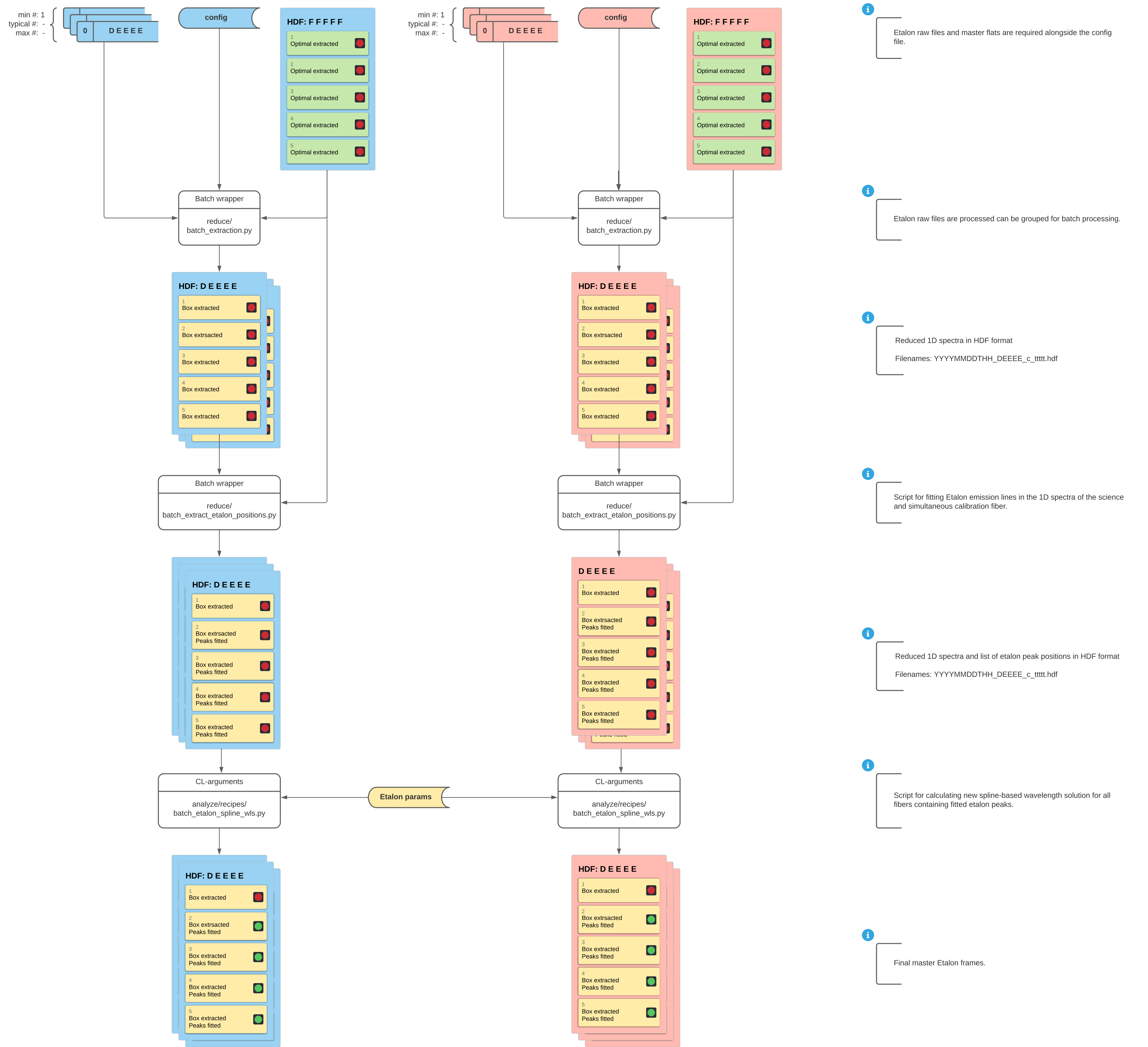
i

Sigma-clipped mean through the stack of images to remove cosmic ray hits and improve SNR. Also produces minimum image for QC as well as some diagnostic output on the console.

i

Filenames: YYYYMMDDTHH_masterdark_mean_DDDDE_c_tttt.fits

Master Etalons



Science Data

